Week 4: Sets

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- Set and elements, subsets.
- Examples of sets (numbers, finite sets, power sets).
- Set operations: union, intersect, complement.
- Compare set identities with logical equivalences.

Definition

- A **set** is an unordered collection of distinct objects, called **elements** or **members** of the set.
- A set is said to **contain** its elements.
- We write $a \in A$ to denote that a is an element of the set A. We say: the element a belongs to the set A.

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- We write $a \in A$ to denote that a is an element of the set A. a in A
- The notation $a \notin A$ denotes that a is not an element of the set A. In logic, it means: ~(a ∈ A)

How to describe a set? list **Roster method**

- The set V of all vowels in English alphabet: $V = \{a, e, i, o, u\}$.
- The set of all non-negative integers less than 10: $S = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\} = \{0, 1, 2, 3, \ldots, 9\}$.
- The set of all odd positive integers less than 100: $S = \{1, 3, 5, \ldots, 99\}$.
- Set of subsets. $A = \{1, 2, 3, 4\}$.
  The set $\mathcal{S}$ of subsets of any two elements in A:

$$\mathcal{S} = \{\{1, 2\}, \{1, 3\}, \{1, 4\}, \{2, 3\}, \{2, 4\}, \{3, 4\}\}.$$

How to describe a set? **Set builder notations**

$$S = \{\text{elements in general form}|\text{conditions}\}$$

or properties

- The set of all odd positive integers less than 100:

$$\begin{aligned} O &= \{n \mid n \text{ is an odd positive integer less than } 100\} \\ &= \{n \mid n \in \mathbb{N},\ n \text{ is odd and } n < 100\} \\ &= \{2k-1|k=1,\ldots,50\}. \end{aligned}$$

a common notation for the set of positive integers

- The set S of all squared numbers $S = \{n^2|n \in \mathbb{N}\}$.
- The set $\mathcal{T}$ of subsets of any two numbers from 1 to 4:

$$\begin{aligned} \mathcal{T} &= \{\{1,2\},\{1,3\},\{1,4\},\{2,3\},\{2,4\},\{3,4\}\} \\ &= \{\{i,j\}|1 \le i < j \le 4 \text{ and } i,j \in \mathbb{N}\}. \end{aligned}$$

Need to write the condition i < j, otherwise i could be equal to j
so subset such as {2,2} = {2} had been collected in T, but it has only one element.
Also otherwise if i ≥ j, then subset such as {2,1} is just the same as {1,2},
which has already been collected in T.

When describing a set:

- The order in which the elements of a set are listed does not matter:

$$\{1,2,3\} = \{3,2,1\} = \{2,1,3\}.$$

- It does not matter if an element of a set is listed more than once:

$$\{1,1,2,2,2,3\} = \{1,2,3\}.$$

Common notations of sets of numbers:

- The set of **natural numbers**

positive integers

$$\mathbb{N} = \{1, 2, 3, \ldots\}.$$

- The set of **integers**

$$\mathbb{Z} = \{\ldots, -3, -2, -1, 0, 1, 2, 3, \ldots\}.$$

- The set of **rational numbers**

fractions / quotients

$$\mathbb{Q} = \{\tfrac{p}{q} | p, q \in \mathbb{Z} \text{ and } q \neq 0\}.$$

- $\mathbb{R}$ = the set of **real numbers**.
- $\mathbb{C}$ = the set of **complex numbers**.

square-root(2), square-root(3), pi, e, …. are irrational numbers. These are real numbers but not rational.

$$\mathbb{C} = \{a + bi | a, b \in \mathbb{R} \text{ and } i^2 = -1\}.$$

We will not touch complex numbers in this course whatsoever.

Definition

Let S be a set.

- If there are exactly n distinct elements in S, we say that S is a **finite** set. (limited)
- n is the **cardinality** of S, and is denoted by $|S|$.
  size
- A set is said to be **infinite** if it is not finite.

Two situations: (1) We can count elements in S, but without end.
(2) We cannot even count them (cannot list them one by one).

Definition

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Finite sets

- **Empty set**: the set that contains no elements, denoted by Ø or { }.
  zero element
  The cardinality I Ø I of the empty set is 0.
- **Singleton set**: a set that contains exactly one element.
  The cardinality of a singleton set is 1.
- **Boolean set**: a set containing two elements $\{0, 1\}$.
  also {1, 2} { T, F }, {1, —1}, {a,b}, ..

In general, if ISI = n, we call S an n-element set.

Intervals

Sets of all the real numbers between two numbers a and b (inclusively or exclusively).

- $[a, b] = \{x \mid a \leq x \leq b$, where x is a real number} closed interval
- $(a, b) = \{x \mid a < x < b$, where x is a real number} open interval
- $(a, b] = \{x \mid a < x \leq b$, where x is a real number}
- $[a, b) = \{x \mid a \leq x < b$, where x is a real number}

Intervals

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- $(a, b] = \{x \mid a < x \leq b\}$
- $[a, b) = \{x \mid a \leq x < b\}$

a can be $-\infty$, and b can be ∞

- Write $(-\infty, b]$, $[a, \infty)$, but never $[-\infty, b]$ or $[a, \infty]$.
- $\mathbb{R} = (-\infty, \infty)$.

What is $\underbrace{(a, a), (a, a], [a, a)}_{\text{emptyset}}$, $\underbrace{[a, a]}_{\text{a singleton set, which is \{a\}}}$?

(a,a) = { x in R I a < x < a } = Ø (no number in this set)

Definition

- Two sets are **equal** if and only if they have the same elements. Namely

$$A = B \quad \Leftrightarrow \quad (\forall x\, (x \in A \leftrightarrow x \in B))$$

- A set A is a **subset** of B if and only if every element of A is also an element of B.

$$A \subseteq B \quad \Leftrightarrow \quad (\forall x \ \ (x \in A \ \rightarrow \ x \in B))$$

possibly equal

- We see that: $\quad A = B \quad \Leftrightarrow \quad (A \subseteq B \text{ and } B \subseteq A).$
- A is a proper subset of B, in notation $A \subset B$, if and only if $A \subseteq B$ and $A \neq B$.

smaller and unequal

Examples.

- $\{1, 3\} \subset \{1, 3, 5\}$.
- $\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}$
- $(a, b) \subset (a, b] \subset [a, b]$, also $(a, b) \subset [a, b) \subset [a, b]$.

  a is in [a,b] on top of (a,b].

  b is in (a,b] on top of (a,b)
- $[a, b] \subseteq [c, d]$ if and only if $c \leq a \leq b \leq d$.

Subset relations

$$A \subseteq B \quad \Leftrightarrow \quad (\forall x \ (x \in A \ \rightarrow \ x \in B))$$

Example. showing subset relations

$$A = \{a | a \text{ is of the form } 6b + 5 \text{ for some } b \in \mathbb{Z}\}.$$
$$B = \{m | m \text{ is of the form } 3n + 2 \text{ for some } n \in \mathbb{Z}\}.$$

A = { 6b + 5 l b in Z } = { … −13 , −7, −1, 5, 11, 17, 23, … }
B = {3n+ 2 l n in Z } = { …, −7, −4, −1, 2, 5, 8, 11, 14, 17, ... }

To show A ⊆B, we take a general element in A, which is 6b + 5 in A,
then we show that this element is in B, that means it can be written as a 3n+2 for some n.

We can write 6b+ 5= 3(2b) + 3+2 = 3(2b+1) + 2, which is in 3n+2 form, with n = 2b+1.
Therefore, 6b+ 5 is in B.

In this example, A is a proper subset of B, because there exists numbers in B which are not in A. Those numbers are the even numbers in B.

Equality of sets

$$A = B \quad \Leftrightarrow \quad (\forall x\,(x \in A \leftrightarrow x \in B))$$

(x ∈A ↔x ∈B) means (x ∈A —> x ∈B) and (x ∈B —> x ∈A)

Example. showing equality of sets

$A = \{m | m$ is of the form $3n + 2$ and $2p + 1$ for some $n, p \in \mathbb{Z}\}$

$B = \{a | a$ is of the form $6b + 5$ for some $b \in \mathbb{Z}\}$.

From the previous page, we see that 6b+ 5 = 3n + 2 where n = (2b+1).
Similarly, we can write 6b+ 5 = 2(3b+2) + 1 = 2p+1, where p = (3b+2).
So any 6b+ 5 is a 3n+ 2 and also a 2p+1, that means: B ⊆A.

To show A ⊆B:

take any number in A, which is in (3n+2)-form and at the same time a (2p+1)-form.
(1) A number in (3n+2)-form, since n can be either 2m or 2m+1, we can write
3(2m) + 2 = 6m+2 where n = 2m, or 3(2m+1)+2 = <u>6m+ 5</u> where n = 2m+1.
(2) A number in (2p+1)-form, since p can be 3m, 3m+1, or 3m+2, we can write
2p + 1 as 6m+1, 6m+ 3, or <u>6m+ 5</u>.

The only form that appears in both (1) and (2) is <u>6m+ 5</u>, which is an element in B.
That means any element in A is also an element in B.

Definition

Given a set S, the **power set** of S is the set of all subsets of the set S, and is denoted by $\mathcal{P}(S)$.

A set is a subset of itself, so S ∈ P(S). The empty set is a subset of any set, so Ø ∈ P(S).

- All the subsets of the set $S = \{a\}$.

The power set P(S) = { Ø, {a} }. The cardinality |P(S)| = 2.

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- All the subsets of the set $S = \{a, b\}$.

  The power set P(S) = { Ø, {a,b}, {a}, {b} }.
  The cardinality IP(S)I = 4.

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- All the subsets of the set $S = \{a, b\}$.

- All the subsets of the set $S = \{a, b, c\}$.

  The power set P(S) = { Ø, {a}, {b}, {c}, {a,b}, {a,c}, {b,c}, S }.
  The cardinality IP(S)I = 8.

Definition

Given a set S, the **power set** of S is the set of all subsets of the set S, and is denoted by $\mathcal{P}(S)$.

- All the subsets of the set $S = \{a\}$.

- All the subsets of the set $S = \{a, b\}$.

- All the subsets of the set $S = \{a, b, c\}$.

If $|S| = n$, then $|\mathcal{P}(S)| =$ (the number of subsets in S) = 2^n.

Definition

Let A and B be sets. The **Cartesian product** of A and B, denoted by $A \times B$, is the set of all ordered pairs (a, b) where $a \in A$, $b \in B$. Namely,

$$A \times B = \{(a, b) \mid a \in A, b \in B\}.$$

In this context, (a,b) is not the open interval.
In Cartesian product, (a,b) is like a vector with two coordinates, like a point (x,y) in a 2D-plane.

Definition

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A x B is not the same as B x A.

$$A \times B = \{(a, b) \mid a \in A, b \in B\}.$$

- If $A = \{1, 2\}$ and $B = \{a, b, c\}$, find

$$A \times B, \quad B \times A, \quad A^2, \quad \text{and} \quad B^2.$$

A x B = { (1, a), (1, b), (1,c), (2,a), (2,b), (2,c) }
B x A = { (a, 1), (a,2) , (b, 1), (b,2), (c, 1), (c, 2) }

A x A = A^2 = { (1,1), (1,2), (2,1), (2,2) }

- Compute $|A \times B|$, $|B \times A|$, $|A^2|$, and $|B^2|$. 6, 6, 4, 9.

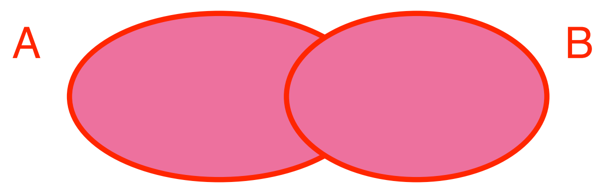
In general, lA x B l = lAl lBl

Set Operations

Definition

The **union** of the sets A and B, denoted by $A \cup B$, is the set that contains those elements that are in A or B.

$$A \cup B = \{x \mid x \in A \lor x \in B\}$$



Set Operations

Definition

The **union** of the sets A and B, denoted by $A \cup B$, is the set that contains those elements that are in A or B.

$$A \cup B = \{x \mid x \in A \ \vee \ x \in B\}$$

Example. If $A = \{1, 3, 5\}$ and $B = \{1, 2, 3\}$, find $A \cup B$.

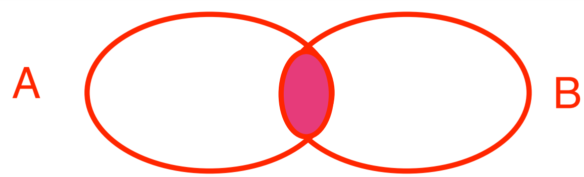
A ∪B = { 1,3,5, 1,2,3 } = { 1, 3, 5, 2 } .

Example. $A = [1, 3]$ and $B = (2, 4)$. A ∪B = [1, 4).

Definition

- The **intersection** of the sets A and B, denoted by $A \cap B$, is the set that contains those elements that are in both A and B.

$$A \cap B = \{x \mid x \in A \ \land \ x \in B\}$$ overlap



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- Two sets are **disjoint** if their intersection is empty: $A \cap B = \emptyset$. no overlap

Definition

- The **intersection** of the sets A and B, denoted by $A \cap B$, is the set that contains those elements that are in both A and B.

$$A \cap B = \{x \mid x \in A \ \wedge \ x \in B\}$$

- Two sets are **disjoint** if their intersection is empty: $A \cap B = \emptyset$.

Example. If $A = \{1, 3, 5\}$ and $B = \{1, 2, 3\}$, find $A \cap B$.

A ∩B = {1, 3}

Example. $A = \{1, 2, 3\}$ and $B = \{6, 8\}$ are disjoint.

There is no common element in both A and B.

Example. Let

$$A = \{x \in \mathbb{N} \mid x \text{ is an odd number less than } 10\},$$
$$B = \{x \in \mathbb{Z} \mid x^2 \leq 20\}.$$

Find $|A \cup B|$ and $|A \cap B|$.

A = { 1, 3, 5, 7, 9}, |A|= 5

B = { −4, −3, −2, −1 , 0, 1, 2, 3, 4}, |B|= 9.

A ∪B = { 1,3,5,7,9, −4, −3, −2, −1, 0, 2, 4} , and |A ∪B| = 12.

A ∩B = { 1, 3}, and |A ∩B| = 2.

Example. Let

$$A = \{x \in \mathbb{N} \mid x \text{ is an odd number less than } 10\},$$
$$B = \{x \in \mathbb{Z} \mid x^2 \leq 20\}.$$

Find $|A \cup B|$ and $|A \cap B|$.

Formula

If A and B are finite sets, then (Inclusion-Exclusion)

$$|A \cup B| = |A| + |B| - |A \cap B|.$$

In the example, $12 = 5 + 9 - 2$.

Definition

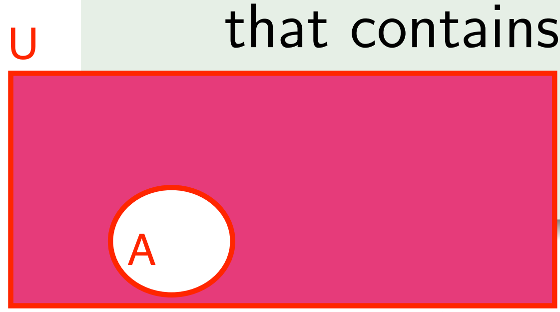
- The **difference** of the set A and set B, denoted by $A \setminus B$, is the set containing elements in A but not in B.

$$A \setminus B = \{x \,|\, x \in A \wedge x \notin B\}$$

- The **universal set** $\mathcal{U}$ is the set that contains all objects under consideration

  Example, discrete math, usually U = Z (integers)
  calculus, U = R (real numbers)
  engineers, U = C (complex numbers)
  in class, U = whiteboard.

- The **complement** of the set A, denoted by $\overline{A}$, is the set that contains all elements in $\mathcal{U}$ but not in A.



$$\overline{A} = \{x \in \mathcal{U} \,|\, x \notin A\} = \mathcal{U} \setminus A$$

Example Let the universal set $\mathcal{U}$ be the set of positive integers, O the set of odd positive integers. Find $\overline{O}$.

U = N

$\overline{\text{O}}$ = the set of even positive integers.

Example Let the universal set $\mathcal{U}$ be the set of positive integers, O the set of odd positive integers. Find $\overline{O}$.

Example Now let $\mathcal{U} = \mathbb{Z}$, and O as above. Find $\overline{O}$.

O = {odd positive integers}

$\overline{O}$ = {even positive integers, zero, or negative integers}

2, 4, 6, 8, … 0 −1, −2, −3, −4, ….

Example Let the universal set $\mathcal{U}$ be the set of positive integers, O the set of odd positive integers. Find $\overline{O}$.

Example Now let $\mathcal{U} = \mathbb{Z}$, and O as above. Find $\overline{O}$.

Example Let the universal set $\mathcal{U}$ be the set of positive real numbers, and let $A = [2, 3)$. Find $\overline{A}$.

$\overline{A}$ = (0, 2) ∪ [3, infinity)

De Morgan's Laws

$$\overline{A \cap B} = \overline{A} \cup \overline{B}, \quad \overline{A \cup B} = \overline{A} \cap \overline{B}.$$

Compare with the De Morgan's laws in logic:

$$\neg(p \wedge q) \equiv \neg p \vee \neg q, \quad \neg(p \vee q) \equiv \neg p \wedge \neg q.$$

Take p: $x \in A$, q: $x \in B$, then $\overline{A \cap B} = \{ x \mid \sim(x \in A \cap B) \}$

$= \{ x \mid \sim(x \in A \wedge x \in B) \} = \{ x \mid \sim(x \in A) \vee \sim(x \in B) \}$ using De Morgan's law in logic

$= \{x \mid x \in \overline{A} \vee$ x in $\overline{B} \} = \overline{A} \cup \overline{B}$

Proving set identities **using set builder notation and logical equivalences.**

Example. Show that $\overline{A \cup B} = \overline{A} \cap \overline{B}$.

Similar work shown on the previous page.